

Hypothesis Testing with E-Values

Chapter 10: Approximate and Asymptotic E-Values

Based on Ramdas & Wang

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Why bother with approximate or asymptotic e-values? I

The problem. So far, an e-variable E for a null $\mathbb{P}$ had to satisfy $\mathbb{E}^{\mathbb{P}}[E] \leq 1$ *exactly*, for every $\mathbb{P}$ in the null, no matter the sample size. That is a strong requirement. In practice:

- We often only know how to construct an e-value that satisfies $\mathbb{E}^{\mathbb{P}}[E] \leq 1$ up to some small, known slack – e.g. because of a plug-in estimate, a truncation, or a numerical approximation.
- Even more commonly, we only know that a statistic *becomes* valid as the sample size $n \rightarrow \infty$ – this is exactly the situation whenever we invoke the Central Limit Theorem (CLT) to justify a Z -test or t -test.

This chapter's job: make these two relaxations – *approximate* (Section 10.1) and *asymptotic* (Section 10.2) – mathematically precise, show they compose well with calibration (converting p-values $\leftrightarrow$ e-values), give the single most important construction (the CLT-based e-value, Section 10.3), and extend everything to *multiple* testing (Section 10.4).

Why bother with approximate or asymptotic e-values? II

Roadmap of the chapter.

- ① **10.1** Approximate p-/e-values: a *fixed*, known error budget (ε, δ) at any single sample size.
- ② **10.2** Asymptotic p-/e-values: a *sequence* of approximate p-/e-values indexed by n , whose error budget $(\varepsilon_n, \delta_n) \rightarrow (0, 0)$ as $n \rightarrow \infty$.
- ③ **10.3** The single most useful asymptotic e-value in practice: built from the CLT, for testing a population mean when the variance is unknown.
- ④ **10.4** The same three ideas (exact / approximate / asymptotic), but now for *compound* e-values used in multiple testing and FDR control (Chapter 9).

Notation you'll need I

Intuition first. An *exact* e-value never lies (in expectation) – but exact constructions are not always available. An *approximate* e-value lies a little, but by a known, bounded amount – you can always account for the slack. An *asymptotic* e-value's lie shrinks to zero only as $n \rightarrow \infty$ – at any *fixed*, finite n you have no numerical guarantee at all, only the promise that the guarantee improves as more data arrives.

Notation you'll need II

Exact vs. approximate vs. asymptotic e-values

Type	Defining condition	What you actually get
Exact	$\mathbb{E}^{\mathbb{P}}[E] \leq 1$ for every $\mathbb{P}$ in the null, at the sample size you have.	A hard guarantee, right now, with zero slack.
Approximate (ε, δ)	$\mathbb{E}^{\mathbb{P}}[E \wedge t] \leq 1 + \varepsilon + \delta t$ for all $t \in \mathbb{R}_+$, at the sample size you have.	A guarantee with a <i>known, bounded</i> error: $\varepsilon \geq 0$ is a multiplicative slack, $\delta \in [0, 1]$ is a probability-of-failure slack. Still holds at any fixed n .
Asymptotic	A <i>sequence</i> $(E^{(n)})_n$ that is $(\varepsilon_n, \delta_n)$ -approximate with $\varepsilon_n(\mathbb{P}) \rightarrow 0$, $\delta_n(\mathbb{P}) \rightarrow 0$ as $n \rightarrow \infty$.	No numerical guarantee at any single finite n – only that the error vanishes in the limit $n \rightarrow \infty$.

Notation you'll need III

Here $E \wedge t := \min(E, t)$; capping E at a threshold t before taking expectations is what makes the definition work even for E that can be infinite (recall E is $[0, \infty]$ -valued).

Notation you'll need IV

Two different kinds of error, ε and δ . Definition 10.1 in the book distinguishes two ways an object can fail to be a genuine p-/e-variable:

ε : a multiplicative value-error

Measures how far the *value* of the random variable is from a genuine p-/e-variable. Example: if $P \sim U(0, 1)$ is a genuine p-variable, then $(1 + \varepsilon)P$ is “ ε -away” – every realization is inflated by the same known factor.

δ : a probability-of-failure error

Measures *how often* (in probability) the object deviates from being a genuine p-/e-variable. Example: if $P \sim U[0, 1]$ and A is an event with $\mathbb{P}(A) = 1 - \delta$, then $P \mathbb{I}_A$ is “ δ -away” – it behaves perfectly on A , and is simply set to 0 (an automatic rejection) with probability δ .

The two error types are asymmetric: δ is the more lenient/forgiving of the two (Proposition 10.5 below makes this precise).

Notation you'll need V

CLT recap (the tool that powers all of Section 10.3). Let $X^1, \dots, X^n$ be i.i.d. real random variables with $\mathbb{E}[X^i] = \mu$ and $\text{var}(X^i) = \sigma^2 \in (0, \infty)$. Write the sample mean $\bar{X}^{(n)} = \frac{1}{n} \sum_{i=1}^n X^i$.

Central Limit Theorem

$$\sqrt{n} \frac{\bar{X}^{(n)} - \mu}{\sigma} \xrightarrow{d} N(0, 1) \quad \text{as } n \rightarrow \infty,$$

i.e. for large n , $\bar{X}^{(n)}$ is *approximately* Gaussian with mean μ and variance σ^2/n : $\bar{X}^{(n)} \approx N(\mu, \sigma^2/n)$. The concrete variance formula $\text{var}(\bar{X}^{(n)}) = \sigma^2/n$ is exactly why more data ($n \rightarrow \infty$) shrinks the spread of $\bar{X}^{(n)}$ around μ to zero.

This is an *asymptotic* statement: it says nothing exact about any fixed n , only that the approximation error vanishes as $n \rightarrow \infty$ – exactly the spirit of “asymptotic” e-values we formalize next.

10.1 Intuition: allowing a known amount of slack

Why not require exactness always? Sometimes we can prove a statistic is *almost* a valid p-/e-variable, with an error we can bound explicitly – e.g. from a plug-in variance estimate, a discretization, or a truncation used for computational reasons. Definition 10.1 packages this “almost valid, with a known bound on how far off” idea into two tuning knobs, ε (value-error) and δ (probability-error), that can be set to 0 to recover the exact definitions.

Definition 10.1: approximate p-variables and e-variables

Fix the null $\mathcal{P}$ and two functions $\varepsilon : \mathcal{P} \rightarrow \mathbb{R}_+$, $\delta : \mathcal{P} \rightarrow [0, 1]$; write $\varepsilon_{\mathbb{P}} = \varepsilon(\mathbb{P})$, $\delta_{\mathbb{P}} = \delta(\mathbb{P})$.

(ε, δ) -approximate p-variable

An $\mathbb{R}_+$ -valued random variable P is (ε, δ) -approximate for $\mathcal{P}$ if

$$\mathbb{P}(P \leq t) \leq (1 + \varepsilon_{\mathbb{P}})t + \delta_{\mathbb{P}} \quad \text{for all } t \in (0, 1), \mathbb{P} \in \mathcal{P}.$$

Here t ranges over candidate significance thresholds; the right side inflates the ideal bound t by a multiplicative factor $1 + \varepsilon_{\mathbb{P}}$ and an additive slack $\delta_{\mathbb{P}}$.

(ε, δ) -approximate e-variable

A $[0, \infty]$ -valued random variable E is (ε, δ) -approximate for $\mathcal{P}$ if

$$\mathbb{E}^{\mathbb{P}}[E \wedge t] \leq 1 + \varepsilon_{\mathbb{P}} + \delta_{\mathbb{P}}t \quad \text{for all } t \in \mathbb{R}_+, \mathbb{P} \in \mathcal{P}.$$

Setting $\varepsilon \equiv 0, \delta \equiv 0$ recovers ordinary p-/e-variables exactly. If $\delta_{\mathbb{P}} = 1$, the definition is vacuous (no constraint at all) – so δ close to 1 buys almost nothing.

Proposition 10.2: an equivalent, more intuitive picture

Intuition. Rather than an abstract inequality, you can picture (ε, δ) -approximation concretely: there is a “good” event A , of probability at least $1 - \delta_{\mathbb{P}}$, on which the object behaves like an honest $(\varepsilon_{\mathbb{P}}, 0)$ -approximate p-/e-variable; outside A (probability $\leq \delta_{\mathbb{P}}$), anything at all can happen.

Proposition 10.2

For every $\mathbb{P} \in \mathcal{P}$ there exists an event A such that:

$$\text{(p-variable)} \quad \mathbb{P}(P \leq t, A) \leq (1 + \varepsilon_{\mathbb{P}})t \quad \forall t \in (0, 1), \quad \mathbb{P}(A) \geq 1 - \delta_{\mathbb{P}};$$

$$\text{(e-variable)} \quad \mathbb{E}^{\mathbb{P}}[E\mathbb{1}_A] \leq 1 + \varepsilon_{\mathbb{P}}, \quad \mathbb{P}(A) \geq 1 - \delta_{\mathbb{P}}.$$

This event-based statement always *implies* the Definition 10.1 inequality; the two are fully *equivalent* whenever every $\mathbb{P} \in \mathcal{P}$ is atomless (so we can construct A exactly at the right probability level).

Calibration still works: Theorem 10.3 and Proposition 10.5

Theorem 10.3 (calibration is preserved)

Using any (p-to-e or e-to-p) calibrator, calibrating an (ε, δ) -approximate p-variable yields an (ε, δ) -approximate e-variable, and vice versa.

This means all the calibration machinery from Chapter 2 (e.g. $f(x) = 1/\sqrt{x}$ or any decreasing density calibrator) carries over verbatim to the approximate setting – you do not need new calibrators.

Proposition 10.5 (δ is more forgiving than ε)

Let $\delta' = (\varepsilon + \delta)/(1 + \varepsilon)$. Any (ε, δ) -approximate p-variable (or e-variable) for $\mathcal{P}$ is automatically also a $(0, \delta')$ -approximate p-variable (or e-variable) for $\mathcal{P}$, with δ' strictly smaller than $\varepsilon + \delta$ whenever $\varepsilon > 0$.

Interpretation: you can always trade a bit of multiplicative slack ε for a (smaller) increase in probability slack δ' – so δ -type approximation is the more general, more lenient notion. This asymmetry is why Section 10.2 below builds two *different* flavors of “asymptotic” out of $\varepsilon_n \rightarrow 0$ alone versus $\varepsilon_n, \delta_n \rightarrow 0$ jointly.

Python: 10.1 – an (ε, δ) -approximate p-value in action

```
import numpy as np
rng = np.random.default_rng(0)

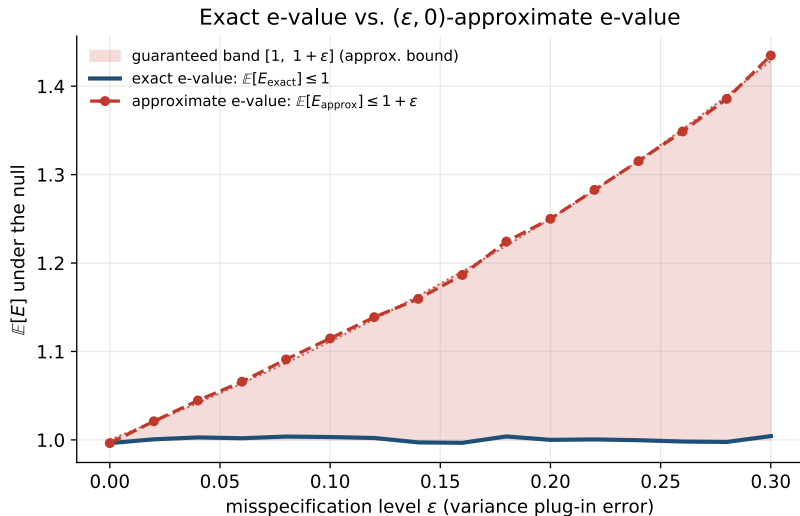
N = 200_000
U = rng.uniform(0, 1, size=N)          # genuine p-variable:  $U \sim \text{Unif}(0,1)$ 

eps, delta = 0.10, 0.05                # known error budget

# Construct an  $(\text{eps}, \text{delta})$ -approximate p-variable "by hand":
# (a) inflate values by  $(1+\text{eps})$  on a "good" event  $A$  of probability  $1-\text{delta}$ ,
# (b) force  $P = 1$  (never falsely reject) on the complement of  $A$ .
A = rng.uniform(0, 1, size=N) <= (1 - delta)
P_tilde = np.where(A, (1 + eps) * U, 1.0)

t = 0.05                               # a candidate significance level
lhs = np.mean(P_tilde <= t)             # empirical  $P(P_{\text{tilde}} \leq t)$ 
rhs = (1 + eps) * t + delta              # Definition 10.1 bound
print(f"P(P_tilde <= {t}) = {lhs:.4f} vs. (1+eps)t + delta = {rhs:.4f}")
# lhs should stay safely below rhs -- the approximate p-value bound holds
```

Figure: exact vs. $(\varepsilon, 0)$ -approximate e-value



Null population $N(0, 1)$, testing $\mathbb{P}_\sigma = \{\mathbb{P} : \mathbb{E}[X] = 0, \text{var}(X) \leq \sigma^2\}$ (Example 10.12). The exact e-value $E_{\text{exact}} = Y^2/\sigma^2$ has $\mathbb{E}[E_{\text{exact}}] \leq 1$ for every misspecification level ε (flat blue line). An approximate e-value

10.2 Intuition: an entire sequence, with vanishing error

The idea. We now observe growing datasets $X^{(1)}, X^{(2)}, \dots$ and compute a statistic $P^{(n)}$ or $E^{(n)}$ from each. Rather than requiring (ε, δ) -approximation with a *fixed*, known (ε, δ) at every n , we only ask that *some* sequence $(\varepsilon_n, \delta_n) \rightarrow (0, 0)$ works – we do not even need to know the rate of convergence, or the value of $(\varepsilon_n, \delta_n)$ at any particular n . This is strictly weaker than approximate validity at each fixed n , and it is exactly the guarantee that CLT-based constructions provide.

Definition 10.6: four flavors of “asymptotic”

Asymptotic e-variables

$(E^{(n)})_{n \in \mathbb{N}}$ is a sequence of *asymptotic* e-variables for $\mathcal{P}$ if $E^{(n)}$ is $(\varepsilon_n, \delta_n)$ -approximate for some $(\varepsilon_n, \delta_n)$ with, for each $\mathbb{P} \in \mathcal{P}$: $\lim_n \varepsilon_n(\mathbb{P}) = \lim_n \delta_n(\mathbb{P}) = 0$.

Flavor	Extra requirement beyond “asymptotic”
Strongly asymptotic	$\delta_n(\mathbb{P}) = 0$ for n large enough (not just $\rightarrow 0$) – the probability-error disappears completely, eventually.
Uniformly asymptotic	$\sup_{\mathbb{P} \in \mathcal{P}} \varepsilon_n(\mathbb{P}) \rightarrow 0$ and $\sup_{\mathbb{P} \in \mathcal{P}} \delta_n(\mathbb{P}) \rightarrow 0$ – convergence (Uniformly, rate does not depend on which null $\mathbb{P}$ is true.
Uniformly strongly asymptotic	Both of the above simultaneously.

strongly) asymptotic p-variables are defined analogously, replacing the e-variable inequality with the p-variable one. If $\mathcal{P}$ is finite, “asymptotic” automatically upgrades to “uniformly asymptotic.”

Proposition 10.7: useful equivalent characterizations I

Intuition first. Part (i) is the workhorse: if a sequence of statistics converges *in distribution* to a genuine p-/e-variable, it is automatically an asymptotic p-/e-variable – this is exactly how the CLT will be used in Section 10.3. Parts (ii)–(iv) restate the definitions in terms of expectations/probabilities, which are often easier to verify directly.

Proposition 10.7

- ① If $(P^{(n)})$ (resp. $(E^{(n)})$) converges in distribution under every $\mathbb{P} \in \mathcal{P}$ to a genuine p-variable (resp. e-variable), it is a sequence of asymptotic p-variables (resp. e-variables).
- ② $(E^{(n)})$ is uniformly strongly asymptotic iff $\limsup_n \sup_{\mathbb{P} \in \mathcal{P}} \mathbb{E}^{\mathbb{P}}[E^{(n)}] \leq 1$; it is strongly asymptotic iff $\limsup_n \mathbb{E}^{\mathbb{P}}[E^{(n)}] \leq 1$ for every fixed $\mathbb{P}$.
- ③ If a sequence of asymptotic e-variables is uniformly integrable (for each $\mathbb{P}$), it is strongly asymptotic.
- ④ $(P^{(n)})$ is uniformly asymptotic iff $\limsup_n \sup_{\mathbb{P} \in \mathcal{P}} \mathbb{P}(P^{(n)} \leq t) \leq t$ for all $t \in (0, 1)$.

Proposition 10.7: useful equivalent characterizations II

Why this matters in practice. Constructions based on convergence in distribution (part (i)) are ubiquitous: Z-tests and t-tests rely on $Z^{(n)} \rightarrow N(0, 1)$; likelihood-ratio and χ^2 -tests rely on $Z^{(n)} \rightarrow \chi^2_\nu$. Whenever $\mathbb{E}[h(Z)] \leq 1$ for the limiting law Z and h is increasing/decreasing/upper semicontinuous, Proposition 10.9 (below) turns this convergence directly into a sequence of asymptotic e-variables $E^{(n)} = h(Z^{(n)})$ – no further work needed. Part (iii) tells us when we get the *stronger* “strongly asymptotic” flavor for free: whenever the sequence is uniformly integrable, which is often checkable via a uniform moment bound (as we do for Theorem 10.11 below).

Proposition 10.8 (calibration, again)

Calibrating a sequence of (uniformly, strongly) asymptotic p-variables – applying a possibly different calibrator to each element – yields a sequence of (uniformly, strongly) asymptotic e-variables, and vice versa.

Proposition 10.9: the general convergence-in-distribution recipe

Proposition 10.9

Suppose a statistic $Z^{(n)} \rightarrow Z$ in distribution under every $\mathbb{P} \in \mathcal{P}$, where Z has a continuous CDF F .

- ① $P^{(n)} = F(Z^{(n)})$ or $1 - F(Z^{(n)})$ defines a sequence of asymptotic p-variables.
- ② If $h : \mathbb{R} \rightarrow [0, \infty)$ is increasing, decreasing, or upper semicontinuous with $\mathbb{E}[h(Z)] \leq 1$, then $E^{(n)} = h(Z^{(n)})$ defines a sequence of asymptotic e-variables; if h is also bounded, the sequence is *strongly* asymptotic.

Three limiting laws that show up constantly:

- $Z \sim N(0, 1)$ – from the CLT (Section 10.3's construction).
- $Z \sim |N(0, 1)|$, the folded normal (distribution of $|X|$, $X \sim N(0, 1)$) – for two-sided tests.
- $Z \sim \chi^2_\nu$ – for likelihood-ratio or chi-square tests.

This single proposition is the engine behind essentially every classical large-sample test being reinterpreted as producing an asymptotic e-value, not just an asymptotic p-value.

10.3 Intuition: turning the Z -test into an e-value

The classical picture. Every introductory statistics course teaches: to test $H_0 : \mathbb{E}[X] = 0$ with unknown variance, compute $Z^{(n)} = \sqrt{n}\bar{X}^{(n)}/S^{(n)}$, and by the CLT this is approximately $N(0, 1)$ under H_0 for large n – giving an (asymptotic) p -value.

The new idea in this section. Instead of stopping at a p -value, we exploit the *exact* moment generating function identity of the standard normal, $\mathbb{E}[\exp(\lambda Z - \lambda^2/2)] = 1$ for $Z \sim N(0, 1)$, to build an e-value directly out of $Z^{(n)}$. Because $Z^{(n)}$ is only *approximately* Gaussian at finite n , the resulting e-value is only *asymptotically* valid – but this is often the *only* e-value construction available for this null (Theorem 10.11's remark), since no exact, nonasymptotic e-variable is known for testing a mean with completely unknown, unbounded variance.

Step-by-step derivation of the CLT-based e-value

Setup. Observe $X^{(n)} = (X^1, \dots, X^n)$, i.i.d. $\sim \mathbb{P}$, testing $\mathcal{P} = \{\mathbb{P} : \mathbb{E}^{\mathbb{P}}[X^i] = 0, \text{ var}^{\mathbb{P}}(X^i) \in (0, \infty)\}$ (mean zero, unknown finite variance).

Step 1: standardize the sample mean

Form $\bar{X}^{(n)} = \frac{1}{n} \sum_i X^i$ and $S^{(n)} = \sqrt{\frac{1}{n} \sum_i (X^i)^2}$ (a plug-in scale, close to the standard deviation under H_0). By the CLT, the law of large numbers, and Slutsky's theorem,

$$Z^{(n)} := \sqrt{n} \frac{\bar{X}^{(n)}}{S^{(n)}} \xrightarrow{d} Z \sim \text{N}(0, 1) \quad \text{as } n \rightarrow \infty.$$

Step 2: use the (approximate) Gaussian likelihood ratio as the e-value

Fix $\lambda \in \mathbb{R}$ (a tuning parameter, chosen in the direction of an anticipated alternative). Define

$$E^{(n)} = \exp\left(\lambda Z^{(n)} - \frac{\lambda^2}{2}\right) = \exp\left(\lambda \frac{\sqrt{n} \bar{X}^{(n)}}{S^{(n)}} - \frac{\lambda^2}{2}\right).$$

This is exactly the likelihood ratio of $\text{N}(\lambda, 1)$ to $\text{N}(0, 1)$, evaluated at $Z^{(n)}$ – and $\mathbb{E}[\exp(\lambda Z - \lambda^2/2)] = 1$ exactly for the limiting $Z \sim \text{N}(0, 1)$

Example 10.10 and Theorem 10.11: the full statement

Example 10.10 / Eq. (10.4)–(10.5)

With $\hat{\sigma}^{(n)} = \sqrt{\frac{1}{n-1} \sum_i (X^i - \bar{X}^{(n)})^2}$ (the usual sample standard deviation), define the one-sided and two-sided e-values

$$E^{(n)} = \exp\left(\lambda \frac{\sqrt{n}\bar{X}^{(n)}}{S^{(n)}} - \frac{\lambda^2}{2}\right), \quad \tilde{E}^{(n)} = \frac{1}{2}e^{\lambda\sqrt{n}\bar{X}^{(n)}/S^{(n)} - \lambda^2/2} + \frac{1}{2}e^{-\lambda\sqrt{n}\bar{X}^{(n)}/S^{(n)} - \lambda^2/2}.$$

Theorem 10.11

$E^{(n)}$ and $\tilde{E}^{(n)}$ (using $S^{(n)}$) are *strongly* asymptotic e-variables for $\mathcal{P}$; if $\hat{\sigma}^{(n)}$ is used instead of $S^{(n)}$, they are asymptotic (but not necessarily strongly). Under any alternative $\mathbb{Q}$ with $\mathbb{E}^{\mathbb{Q}}[X^i] > 0$ (resp. $\neq 0$) and finite variance, and $\lambda > 0$: $E^{(n)}$ (resp. $\tilde{E}^{(n)}$) $\rightarrow \infty$ almost surely – so the test has power 1 asymptotically.

Where “strongly” comes from. Convergence in distribution alone only gives “asymptotic.” Upgrading to *strongly* asymptotic (Proposition 10.7(iii)) requires showing $(E^{(n)})$ is uniformly integrable – proved here using a moment bound of Giné et al. [1997] for self-normalized sums. **Bonus:** mixing (10.4) over λ against any distribution G with compact support, $\int \exp(\lambda\sqrt{n}\bar{X}^{(n)}/S^{(n)} - \lambda^2/2) dG(\lambda)$, remains a

Contrast: Example 10.12, when an *exact* e-value does exist

Why the contrast matters. Section 10.3's construction is needed precisely because, for $\mathcal{P} = \{\mathbb{P} : \mathbb{E}[X] = 0, \text{var}(X) \in (0, \infty)\}$ (variance completely unknown, unbounded), no nonasymptotic e-variable is known (other than the trivial constant 1). But if we are willing to *additionally assume* a known upper bound on the variance, exact e-values become available immediately.

Example 10.12: testing the mean with *bounded* variance

For $\mathcal{P}_\sigma = \{\mathbb{P} \in \mathcal{M}_1 : \mathbb{E}^\mathbb{P}[X] = 0, \text{var}^\mathbb{P}(X) \leq \sigma^2\}$ with $\sigma > 0$ known, the following are *exact* e-variables for $\mathcal{P}_\sigma$ (no asymptotics needed, valid for every $n \geq 1$):

$$\frac{X^2}{\sigma^2}, \quad \exp\left(\lambda X - \frac{\lambda^2}{6}(X^2 + 2\sigma^2)\right), \quad \exp\left(\phi(\lambda X) - \frac{\lambda^2}{2}\sigma^2\right),$$

where $\phi(x) = \log(1 + x + x^2/2)$ for $x \geq 0$ and $\phi(x) = -\log(1 - x + x^2/2)$ for $x < 0$.

Takeaway: asymptotic e-values are a fallback for when exact constructions are unavailable – whenever a genuinely exact e-value exists (e.g. under a known variance bound), prefer it; it carries a guarantee at every n , not just in the limit.

Running numerical example: centered Exponential data, $n = 30, 100, 1000$

Setup. Let $Y^i \stackrel{\text{iid}}{\sim} \text{Exp}(\text{rate} = 1)$ and set $X^i = Y^i - 1$, so $\mathbb{E}[X^i] = 0$, $\text{var}(X^i) = 1$ – but X^i is strongly right-skewed (skewness = 2), i.e. genuinely *non-Gaussian*. We compute $E^{(n)} = \exp(\lambda \sqrt{n} \bar{X}^{(n)} / S^{(n)} - \lambda^2 / 2)$ with $\lambda = 1$ on one realized dataset at each n :

Actual computed values (single simulated dataset, seed fixed)

n	$\bar{X}^{(n)}$	$S^{(n)}$	$\sqrt{n} \bar{X}^{(n)} / S^{(n)}$	$E^{(n)}$
30	-0.0807	0.8888	-0.4970	0.3690
100	0.0735	1.0686	0.6878	1.2066
1000	-0.0092	0.9690	-0.3009	0.4489

Each realized $E^{(n)}$ fluctuates around 1 (as it must, since $\mathbb{E}[E^{(n)}] \approx 1$ under H_0) – it is *not* monotonically approaching any particular value dataset-by-dataset. What *does* improve systematically with n is the **accuracy of the guarantee itself**, which the next slide quantifies.

The asymptotic guarantee gets tighter as n grows

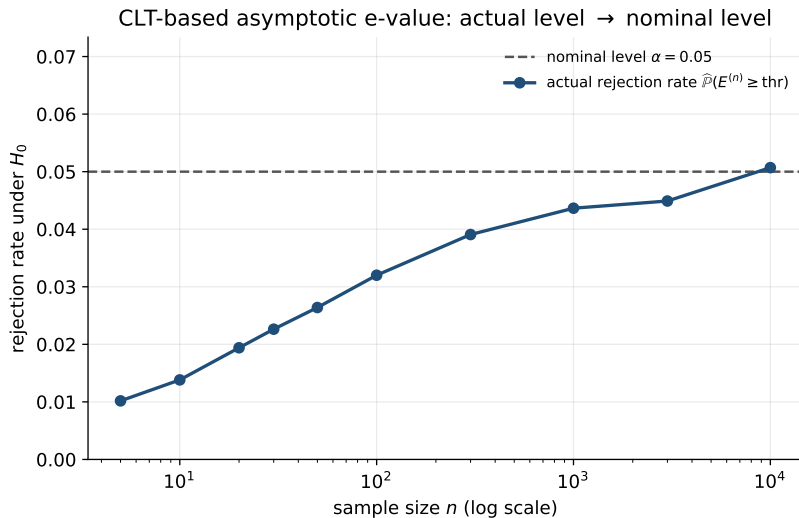
What to check. Fix a nominal level $\alpha = 0.05$ and reject H_0 when $E^{(n)} \geq \exp(\lambda z_{1-\alpha} - \lambda^2/2)$ (equivalent to the classical rule $Z^{(n)} \geq z_{1-\alpha}$, $z_{1-\alpha} = \Phi^{-1}(1 - \alpha)$). Under the CLT, this rejection probability $\rightarrow \alpha$ *exactly* as $n \rightarrow \infty$; the gap at finite n is entirely due to the non-Gaussianity of the centered-Exponential population. Simulating $E^{(n)}$ under H_0 (200,000+ Monte Carlo replicates per n , $\lambda = 1$):

Actual simulated rejection rate vs. nominal $\alpha = 0.05$

n	5	30	100	1000	3000	10000
$\hat{\mathbb{P}}(\text{reject})$	0.0102	0.0226	0.0320	0.0437	0.0449	0.0507

The realized error is $|\hat{\mathbb{P}}(\text{reject}) - 0.05|$: about 0.040 at $n = 5$, shrinking to about 0.001 at $n = 10000$ – exactly the promise of Definition 10.6: $\varepsilon_n, \delta_n \rightarrow 0$, with *no* claim about the rate or the value at any fixed n .

Figure: actual level converges to the nominal level as n grows



Non-Gaussian null population (centered Exponential, skewness 2). The e-value's actual rejection rate under H_0 (solid) starts well below the target $\alpha = 0.05$ at small n and climbs toward it as n grows, illustrating that the

Python: 10.3 – the CLT-based e-value, end to end

```
import numpy as np
from scipy.stats import norm

def clt_evalue(x, lam):
    """ $E^{(n)} = \exp(\lambda * \sqrt{n} * \bar{x} / S - \lambda^2/2)$ , Eq. (10.4)."""
    n = x.shape[-1]
    xbar = x.mean(axis=-1)
    S = np.sqrt((x**2).mean(axis=-1)) # plug-in scale (10.3)
    return np.exp(lam * np.sqrt(n) * xbar / S - lam**2 / 2.0)

rng = np.random.default_rng(7)
lam, alpha = 1.0, 0.05

for n in [30, 100, 1000]:
    Y = rng.exponential(scale=1.0, size=n) # non-Gaussian population
    X = Y - 1.0 # centered:  $E[X] = 0$ 
    E_val = clt_evalue(X, lam)
    print(f"n={n:5d}   $E^{(n)} = \{E\_val:.4f\}")$ 

# Asymptotically-exact rejection threshold:  $E^{(n)} \geq thr \iff Z^{(n)} \geq z_{1-\alpha}$ 
c = norm.ppf(1 - alpha)
threshold = np.exp(lam * c - lam**2 / 2.0)
print(f"reject when  $E^{(n)} \geq \{threshold:.4f\}$  (asymptotically exact level  $\{alpha\})")$ 
```

10.4 Intuition: doing all of the above for MANY hypotheses at once

Chapter 9 introduced *compound* e-variables $E_1, \dots, E_K$ for hypotheses $\mathcal{P}_1, \dots, \mathcal{P}_K$, satisfying an aggregate bound $\sum_{k: \mathbb{P} \in \mathcal{P}_k} \mathbb{E}^{\mathbb{P}}[E_k] \leq K$ (rather than each E_k individually being ≤ 1 in expectation – some can “borrow” slack from others). This section repeats the exact/approximate/asymptotic hierarchy of 10.1–10.3, but for these *aggregate*, multiple-testing objects – because e-BH's FDR control (Chapter 9) only ever needs the aggregate compound bound, everything here plugs directly into FDR-controlling procedures.

10.4.1: exact compound p-variables

Definition 10.13: compound p-variables

$P_1, \dots, P_K$ are *compound p-variables* for $(\mathcal{P}_1, \dots, \mathcal{P}_K)$ under $\mathcal{P}$ if

$$\sum_{k: \mathbb{P} \in \mathcal{P}_k} \mathbb{P}(P_k \leq t) \leq Kt \quad \text{for all } t \in (0, 1), \mathbb{P} \in \mathcal{P}.$$

Theorem 10.14 (calibration, compound version)

Calibrating compound p-variables (with any p-to-e calibrator, the *same* one for every k) yields compound e-variables, and vice versa.

Two illustrative examples. (Ex. 10.15) *Weighted p-values*: given weights $w_k \geq 0$ with $\sum_k w_k \leq K$ and ordinary p-variables P'_k , $\tilde{P}_k = P'_k/w_k$ are compound p-variables. (Ex. 10.16) *Combining*: if P_k is an ordinary p-variable for $\mathcal{P}_k$ and $E_1, \dots, E_K$ are compound e-variables independent of $(P_1, \dots, P_K)$, then P_k/E_k are compound p-variables.

10.4.2: approximate compound p/e-values

Definition 10.17: (ε, δ) -approximate compound e-variables

$E_1, \dots, E_K$ are (ε, δ) -approximate compound e-variables for $(\mathcal{P}_1, \dots, \mathcal{P}_K)$ under $\mathcal{P}$ if, for every $\mathbb{P} \in \mathcal{P}$, there is an event A with $\mathbb{P}(A) \geq 1 - \delta_{\mathbb{P}}$ and

$$\sum_{k: \mathbb{P} \in \mathcal{P}_k} \mathbb{E}^{\mathbb{P}}[E_k \mathbb{1}_{A^c}] \leq K(1 + \varepsilon_{\mathbb{P}}).$$

(Compound p-variables are defined analogously with $\sum_k \mathbb{P}(P_k \leq t, A) \leq K(1 + \varepsilon_{\mathbb{P}})t$.)

Proposition 10.18 (building compound approximations from per-hypothesis ones)

If E_k is $(\varepsilon_k, \delta_k)$ -approximate for $\mathcal{P}_k$ for each k , then $E_1, \dots, E_K$ are (ε, δ) -approximate compound e-variables with $\varepsilon(\mathbb{P}) = \frac{1}{K} \sum_{k: \mathbb{P} \in \mathcal{P}_k} \varepsilon_k(\mathbb{P})$ and $\delta(\mathbb{P}) = \sum_{k: \mathbb{P} \in \mathcal{P}_k} \delta_k(\mathbb{P})$.

So per-hypothesis approximation errors *average* (for ε) but *add up* (for δ) when combined into the compound guarantee – another instance of δ -type slack being more costly to control across many hypotheses. Proposition 10.20 confirms calibration is again preserved in this approximate, compound setting.

10.4.3: asymptotic compound p/e-values and FDR control

Triangular-array asymptotics. We now let *both* $n = n(m)$ (sample size) and $K = K(m)$ (number of hypotheses) vary with an index $m \rightarrow \infty$, requiring $\max\{n, K\} \rightarrow \infty$ – covering Z/t-test-style asymptotics ($n \rightarrow \infty$), empirical-Bayes-style asymptotics ($K \rightarrow \infty$), or both at once.

Definition 10.22: asymptotic compound e-variables

A sequence of tuples $(E_1^{(n(m))}, \dots, E_{K(m)}^{(n(m))})_m$ is *asymptotic compound e-variables* if each tuple is $(\varepsilon_m, \delta_m)$ -approximate compound for some $(\varepsilon_m, \delta_m)$ with $\varepsilon_m(\mathbb{P}), \delta_m(\mathbb{P}) \rightarrow 0$ for each $\mathbb{P} \in \mathcal{P}$ (and *strongly* asymptotic if $\delta_m(\mathbb{P}) = 0$ eventually).

Definition 10.23 / Theorem 10.24: FDR control transfers

A procedure $\mathcal{D}$ has (ε, δ) -approximate FDR control at level α if $\mathbb{E}^{\mathbb{P}}[F_{\mathcal{D}}/R_{\mathcal{D}} \cdot \mathbb{1}_A] \leq \alpha(1 + \varepsilon_{\mathbb{P}})$ on an event A with $\mathbb{P}(A) \geq 1 - \delta_{\mathbb{P}}$. Applying the e-BH procedure (Chapter 9) to **(i)** exact, **(ii)** (ε, δ) -approximate, or **(iii)** asymptotic compound e-variables yields FDR control at level α , $\alpha(1 + \varepsilon) + \delta$, or asymptotic level α , respectively.

10.4.4: the simultaneous t -test sequence model I

Intuition. This is the multi-hypothesis analogue of Section 10.3: K independent groups of data, n i.i.d. observations per group, each group testing its own mean-zero null

$\mathcal{P}_k = \{\mathbb{P} : \mathbb{E}^{\mathbb{P}_k}[X_k^i] = 0, \text{ var}^{\mathbb{P}_k}(X_k^i) \in (0, \infty)\}$. As in (10.3), define per-group $\bar{X}_k^{(n)}, S_k^{(n)}, \hat{\sigma}_k^{(n)}$.

The oracle construction (uses the unknown truth)

Since $\mathbb{E}^{\mathbb{P}_k}[(S_k^{(n)})^2] = \sigma_k^2(\mathbb{P})$ under $\mathcal{P}_k$,

$$\tilde{E}_k^{(n)}(\mathbb{P}) = \frac{K (S_k^{(n)})^2}{\sum_{j:\mathbb{P} \in \mathcal{P}_j} \sigma_j^2(\mathbb{P})} \quad \text{satisfies} \quad \sum_{k:\mathbb{P} \in \mathcal{P}_k} \mathbb{E}^{\mathbb{P}}[\tilde{E}_k^{(n)}(\mathbb{P})] = K.$$

Not usable directly: it depends on the unknown true variances $\sigma_j^2(\mathbb{P})$.

10.4.4: the simultaneous t -test sequence model II

The practical, plug-in construction

Replace the unknown $\sum_j \sigma_j^2(\mathbb{P})$ by the estimable $\sum_j (\hat{\sigma}_j^{(n)})^2$:

$$E_k^{(n)} = \frac{K (S_k^{(n)})^2}{\sum_{j \in \mathcal{K}} (\hat{\sigma}_j^{(n)})^2}, \quad k \in \mathcal{K}.$$

Proposition 10.25

Under mild moment conditions ($\mathcal{P}^B$: bounded $(2 + 2\delta)$ -moments, variance bounded away from $0, \infty$), $(E_k^{(n)})$ is a sequence of asymptotic compound e-variables. If additionally the null marginals are exactly Gaussian with bounded variance ($\mathcal{P}^N$), the sequence is *strongly* asymptotic.

Why it works, in one line. Both numerator sum $n \sum_k (S_k^{(n)})^2 / \sigma_k^2$ and denominator sum $n \sum_k (\hat{\sigma}_k^{(n)})^2 / \sigma_k^2$ concentrate around nK_0 (a $\chi_{nK_0}^2$ -type average), so their ratio $\rightarrow 1$ in probability as $\max\{n, K\} \rightarrow \infty$ – exactly Proposition 10.7(i)'s convergence-in-distribution recipe, applied group-wise.

Python: 10.4 – the simultaneous t -test sequence model

```
import numpy as np
rng = np.random.default_rng(3)

K, n = 40, 50                                # K groups, n obs. per group
# Simulate all K groups under their own null: mean 0, heterogeneous variances
true_sigmas = rng.uniform(0.5, 2.0, size=K)
X = rng.normal(loc=0.0, scale=true_sigmas[:, None], size=(K, n))    # (K, n)

Xbar = X.mean(axis=1)
S2 = (X**2).mean(axis=1)                      #  $S_k^{(n)}$  squared
sigma_hat2 = X.var(axis=1, ddof=1)           #  $\hat{\sigma}_k^2$ 

E = K * S2 / sigma_hat2.sum()                # Proposition 10.25's  $E_k^{(n)}$ 

print("sum_k E[E_k] should concentrate near K =", K)
print("empirical sum of E_k this draw:", E.sum().round(3))

# Repeat many times to see the compound bound sum_k E[E_k] <= K hold on average
reps = np.array([
    (lambda Xr: K * (Xr**2).mean(axis=1) / Xr.var(axis=1, ddof=1).sum())(
        rng.normal(0.0, true_sigmas[:, None], size=(K, n)))
    for _ in range(2000)
])
print("Monte Carlo mean of sum_k E_k across repetitions:", reps.sum(axis=1).mean().round(3))
```

Chapter 10 summary

- **Exact vs. approximate vs. asymptotic** (Section 10.1): an exact e-value has $\mathbb{E}[E] \leq 1$ always; an (ε, δ) -approximate e-value has $\mathbb{E}[E \wedge t] \leq 1 + \varepsilon + \delta t$, a *known, bounded* slack at any fixed n ; an asymptotic e-value is a sequence whose slack $(\varepsilon_n, \delta_n) \rightarrow (0, 0)$, with no guarantee at any single finite n .
- **Calibration is preserved throughout** (Theorems 10.3, Prop. 10.8, 10.14, 10.20): p-to-e and e-to-p calibrators keep working, unchanged, in the approximate/asymptotic/compound settings.
- **The CLT-based e-value** (Section 10.3): standardize $Z^{(n)} = \sqrt{n}\bar{X}^{(n)}/S^{(n)}$, plug into the exact Gaussian MGF identity $\exp(\lambda Z - \lambda^2/2)$, and get a *strongly asymptotic* e-value for testing a mean with completely unknown variance – often the *only* available construction, versus the exact e-values (Example 10.12) available when the variance is known to be bounded.
- **Compound versions** (Section 10.4): the same hierarchy (exact $\rightarrow$ approximate $\rightarrow$ asymptotic) applies to compound e-variables used for FDR control; e-BH's FDR guarantee degrades gracefully from α (exact) to $\alpha(1 + \varepsilon) + \delta$ (approximate) to asymptotic level α (asymptotic), and the simultaneous t -test sequence model (10.4.4) is the multi-hypothesis analogue of Section 10.3's construction.

References

Based on Chapter 10 of A. Ramdas and R. Wang, *Hypothesis Testing with E-values*, arXiv:2410.23614. Materials are largely based on Ignatiadis et al. [2024a]; $(\varepsilon, 0)$ -approximate e-variables (without this terminology) appear in Wang and Ramdas [2022, Sec. 6.2]; the standard limit-law examples of Section 10.2.1 are classical (Van der Vaart [1998], Lehmann and Romano [2005]); Example 10.12 is from Wang and Ramdas [2023]; compound p-variables were introduced (under a different name) by Armstrong [2022] and further studied by Ignatiadis and Sen [2025]; approximate compound e-variables appear implicitly in Ignatiadis et al. [2024b]; a related notion of approximate FDR control was used as a proof technique by Blanchard and Roquain [2008]; Section 10.4.4 follows Ignatiadis et al. [2024a,b].